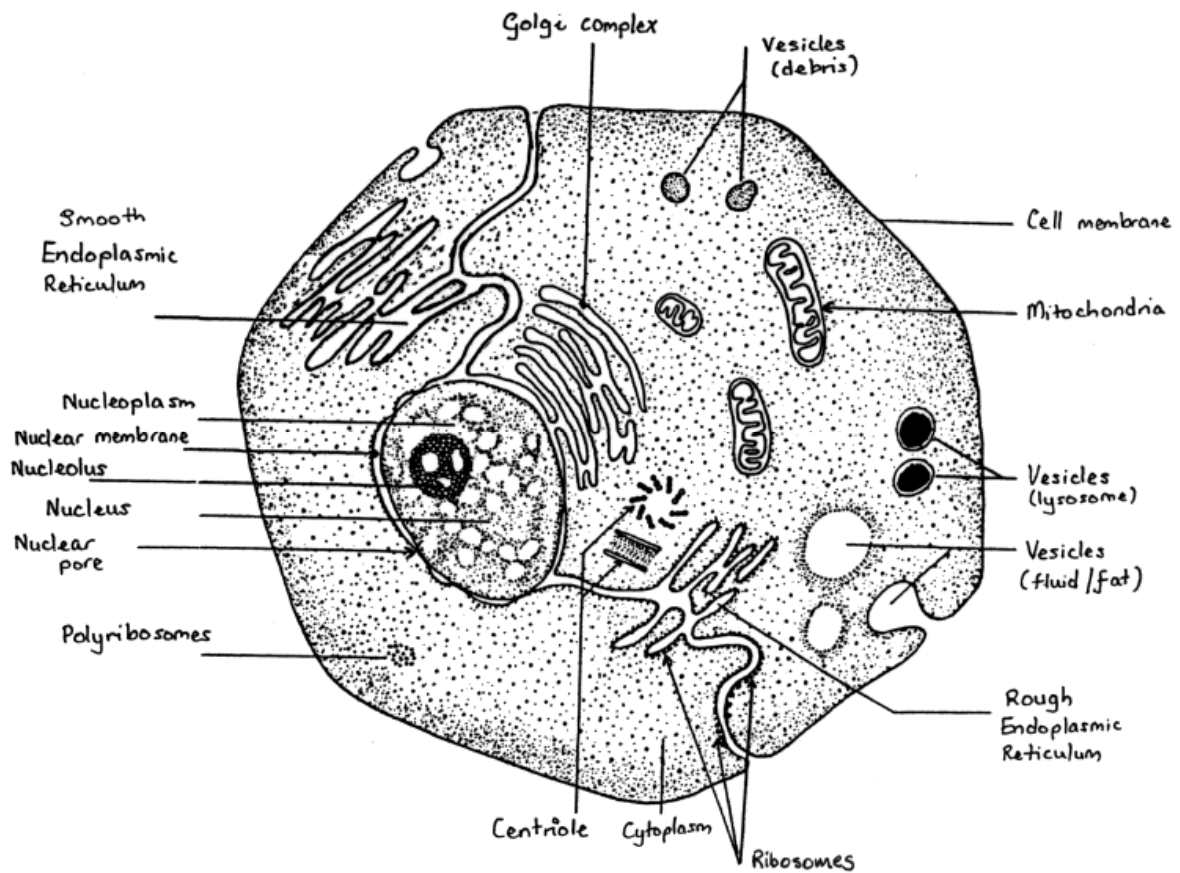


## THE CELL



### Year 10 Information

#### ***Nucleus***

- Controls all cell functions and activity.
- Contains the genetic material (DNA).

#### ***Cytoplasm***

- Carries on the basic processes for the whole cell.

#### ***Cell Membrane***

- Also called the ***plasma membrane***.
- Semi-permeable (has “holes” in it) which allows the entry of food and  $O_2$  and exit of waste products from respiration.

#### ***Cell Wall***

- Provides strength for plant cells (made of cellulose).
- Doesn't exist in animal cells.

## Information For Upper School

|                              |                                                                                                                                                                                                                                                                                                            |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Nucleolus</i>             | <ul style="list-style-type: none"><li>• Produces ribosomes.</li></ul>                                                                                                                                                                                                                                      |
| <i>Ribosomes</i>             | <ul style="list-style-type: none"><li>• Produce proteins for the cell to use.</li></ul>                                                                                                                                                                                                                    |
| <i>Mitochondria</i>          | <ul style="list-style-type: none"><li>• Produce energy for the cell by respiration.</li></ul>                                                                                                                                                                                                              |
| <i>Endoplasmic Reticulum</i> | <ul style="list-style-type: none"><li>• Tubular structure that transports cell products.</li><li>• <b><i>Smooth E.R.</i></b> (without ribosomes) - transports protein for the cell.</li><li>• <b><i>Rough E.R.</i></b> (with ribosomes) - transports protein for <b><i>export</i></b> from cell.</li></ul> |
| <i>Golgi Complex</i>         | <ul style="list-style-type: none"><li>• Packages wastes for export from the cell.</li></ul>                                                                                                                                                                                                                |
| <i>Vesicles</i>              | <ul style="list-style-type: none"><li>• Packages produced by the Golgi complex.</li></ul>                                                                                                                                                                                                                  |
| <i>Centrioles</i>            | <ul style="list-style-type: none"><li>• Produce spindles for chromosome separation in cell division.</li></ul>                                                                                                                                                                                             |